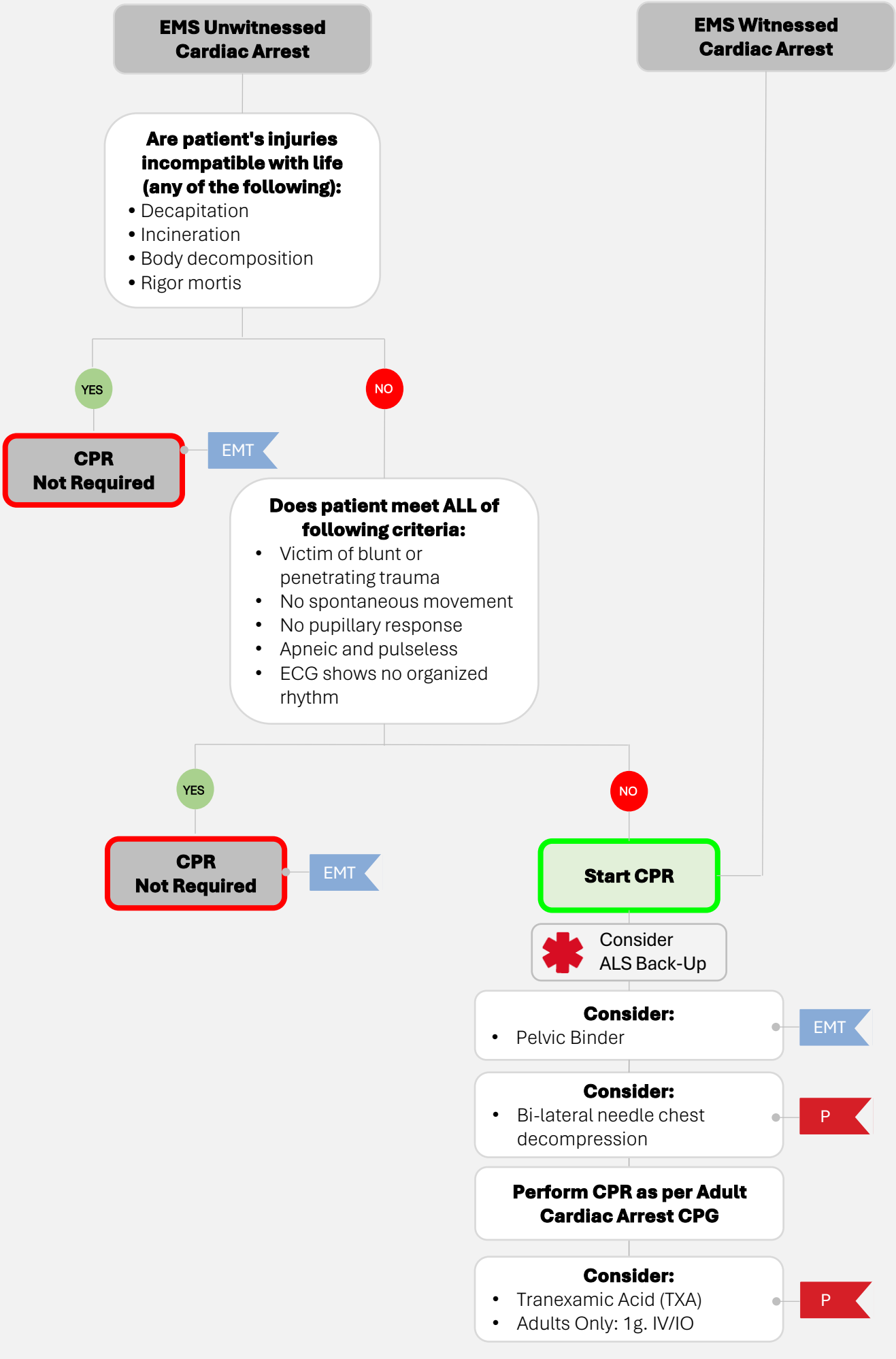
Cardiac Arrest in Trauma



Patient Care Goals

1. Rapidly identify and manage traumatic cardiac arrest.
2. Perform high-quality resuscitation (CPR) and specific interventions related to traumatic arrest to maximize survival.
3. Ensure safe transport to a trauma center.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Adult and pediatric patients presenting with traumatic cardiac arrest, including:
    - Unresponsiveness with no signs of life (e.g., no pulse, no breathing, or agonal gasps) following significant trauma.

Exclusion Criteria

- Non-traumatic cardiac arrest patients.

Key Patient Assessment Points

1. **Traumatic Cardiac Arrest Confirmation**
    - Perform Primary Survey as per Universal Care CPG.
        - Verify unresponsiveness by assessing for no response to verbal or painful stimuli (AVPU).
        - Check for absence of pulse (carotid or femoral) for 5-10 seconds.
        - Confirm absence of normal breathing or presence of agonal gasps for 5-10 seconds.
2. **Trauma Mechanism Assessment**
    - Identify the mechanism of injury (e.g., blunt vs. penetrating, location of trauma such as chest, abdomen, or head).
    - Assess for high-risk injuries likely causing arrest (e.g., tension pneumothorax, massive hemothorax, or cardiac tamponade).
3. **Signs of Irreversible Death**
    - In the unwitnessed traumatic cardiac arrest patient, identify signs of irreversible death, including:
        - Decapitation
        - Incineration
        - Body decomposition
        - Rigor mortis
4. **Futile Resuscitation in Trauma Assessment**
    - In the unwitnessed traumatic cardiac arrest patient, assess for signs which indicate that resuscitation is not indicated. Including:
        - Victim of blunt or penetrating trauma
        - No spontaneous movement
        - No pupillary response
        - Apneic and pulseless
        - ECG shows no organized rhythm
    - All signs must be present to not start resuscitation (CPR)
5. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
        - All patients cardiac arrest are Red Priority.

Patient Management & Interventions

1. **Deciding to Start CPR**
    - **Witnessed Cardiac Arrest**
        - If the Cardiac Arrest was witnessed by ambulance staff, CPR should be initiated.
    - **Unwitnessed Cardiac Arrest**
        - **Signs of Irreversible Death Assessment**:
            - If any of the signs below are present, resuscitation (CPR) is not required. Including:

        - Decapitation
        - Incineration
        - Body decomposition
        - Rigor mortis
    - **Futile Resuscitation in Trauma Assessment:**
        - If all of the below signs are present, resuscitation (CPR) is not required, including:
            - Victim of blunt or penetrating trauma
            - No spontaneous movement (Unresponsive on AVPU scale)
            - No pupillary response (Pupils non-reactive to light).
            - Apneic and pulseless
            - ECG shows no organized rhythm (asystole).

2. **Request for ALS Back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
    - If resuscitation is started ALS back-up should be considered.
    - Transfer to hospital should be prioritized, do not wait on scene, meet the ALS unit en route.
3. **Perform Resuscitation (CPR)**
    - Perform resuscitation as per Adult or Pediatric Cardiac Arrest CPG.
4. **Consider Specific Cardiac Arrest in Trauma Interventions**
    - Consider placing a pelvic binder performing bi-lateral chest needle decompression.
        - Indications:
            - Where the cause of the arrest is not clear (such as in polytrauma in a road traffic accident) application of both the pelvic binder and performing bi-lateral chest needle decompression should be considered.
        - Contraindications:
            - Where there is a clear injury which will not result in pelvic fracture or pneumothorax (such as arterial limb bleeding, or isolated traumatic brain injury).
    - The **pelvic binder should be placed before chest compressions start** – the time *permitted to place the pelvic binder is limited to 30 seconds after no pulse is felt.
    - Active external bleeding should be identified and immediately controlled using direct pressure, wound packing, or tourniquets as appropriate.
        - Uncontrolled hemorrhage is a common reversible cause of traumatic cardiac arrest and must be addressed rapidly.

Patient Safety Considerations

- Resuscitation (CPR) should be initiated in cases when the Mechanism of Injury does not correlate with the clinical condition, suggesting a non-traumatic cause of the arrest.

Clinical Insights & Additional Guidance

- For guidance on stopping or transferring CPR cases in trauma, see the Starting, Stopping or Transferring CPR CPG.
- **Tranexamic Acid (TXA) in Traumatic Cardiac Arrest**
    - General Guidance:
        - TXA is not routinely indicated in patients who are already in cardiac arrest unless there is a high suspicion that the arrest is due to massive hemorrhage and ROSC (Return of Spontaneous Circulation) is achievable.
    - Indications for TXA in Traumatic Cardiac Arrest:
        - If the traumatic cardiac arrest is witnessed, and/
        - There are signs of massive hemorrhage (e.g., penetrating trauma, major limb injury, severe pelvic fracture).
        - TXA 1g IV/IO can be administered as early as possible, ideally within the first 3 hours of injury.
    - Contraindications of TXA in Traumatic Cardiac Arrest:
        - In unwitnessed arrest with prolonged downtime, or when death is clearly non-survivable (e.g., massive head injury, decapitation, hemicorporectomy).
        - If hemorrhage is not suspected to be the cause of arrest (e.g., isolated TBI).

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Mechanism of injury (blunt, penetrating), suspected traumatic cardiac arrest.
- Injuries / Information: Injury pattern noted, signs of irreversible death, resuscitation criteria met or unmet.
- Signs: Initial rhythm, unresponsiveness, pulse and breathing status, GCS if regained, pupil response.
- Treatment:
    - CPR and trauma-specific interventions: pelvic binder placement, bilateral chest decompression, any methods used to stop active bleeding.
    - IV/IO access, airway management, medications if applicable.
- Allergies: Known drug or environmental allergies, if obtainable.
- Medications: Pre-arrest medications if known.
- Background: History if available from bystanders, caregivers, or ID.
- Other: ECG rhythm strips, time CPR started/stopped, transfer or termination rationale.

Key Performance Indicators

1. Immediately initiate CPR for traumatic cardiac arrests witnessed by EMS>
2. Request for back up (where appropriate) for all traumatic cardiac arrests.

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